MasterControl QAL Performance Dataset

Student Analysis Guide

📌 OVERVIEW

You'll be analyzing **16,644 sales leads** from MasterControl to help them improve their sales targeting strategy. This document provides the context and dataset information you need to get started.

🏢 COMPANY BACKGROUND

About MasterControl

MasterControl provides quality and manufacturing management software to life sciences companies (pharmaceutical, biotech, and medical device manufacturers).

Their Products

- **Qx (Quality solutions)**: Established product line with 20+ years in market
  - Current performance: **19.7% conversion rate** ✅
- **Mx (Manufacturing solutions)**: Newer product launched ~4 years ago
  - Current performance: **12.7% conversion rate** ⚠️

The Business Challenge

The Mx product is significantly underperforming compared to Qx. MasterControl believes they may not be targeting the right types of companies or the right people within those companies.

🎯 YOUR RESEARCH OBJECTIVES

Your analysis should help MasterControl answer these questions:

Level 1: Account Analysis

Which types of accounts are best prospects for Mx?

- Industries (pharma vs. medical device vs. non-life science)

- Manufacturing models (in-house vs. contract manufacturing vs. services/consulting)
- Site functions (what they actually manufacture)
- Company size and location

Level 2: Title Analysis

Which job titles convert best within those account types?

- Seniority levels (C-suite, VP, Director, Manager)
- Functional areas (Quality, Operations, Manufacturing, Regulatory)

Level 3: Cross-Sell Opportunities

Which accounts and titles are good for BOTH products?

- Cross-sell opportunities
- Universal high-converting profiles

📊 DATASET INFORMATION

Basic Statistics

- **Total Records**: 16,644 qualified sales leads (QALs)
- **Time Period**: January 1, 2024 - December 31, 2025 (2 complete years)
- **Unique Contacts**: 15,371 people
- **Total Columns**: 14 fields
- **File Name**: `Mastercontrol QAL Performance.csv`

Success Metrics

- **Overall**: 18.0% of leads progressed to next sales stage (2,988 leads)
- **Qx Product**: 19.7% progression rate (2,459 of 12,466 leads)
- **Mx Product**: 12.7% progression rate (529 of 4,178 leads)

Performance Gap: Qx outperforms Mx by 55% (7.0 percentage points) - this is what you're trying to understand and improve.

A lead "progressed" if it moved to SQL (Sales Qualified Lead), SQO (Sales Qualified Opportunity), or Won stages.

📋 DETAILED COLUMN DESCRIPTIONS

🎯 TARGET VARIABLE

`next_stage__c` - Lead Outcome

Data Type: Text (Categorical)

Purpose: This is your dependent variable - it captures whether a sales lead progressed beyond the initial qualification stage.

Missing Values: 394 records (2.37%) - these represent leads still being evaluated

8 Unique Values with Distribution:

- **Recycled**: 12,019 (72.2%) - returned to marketing nurture [NOT progressed]
- **SQL**: 2,694 (16.2%) - progressed to sales [PROGRESSED ✅]
- **Exclude**: 1,219 (7.3%) - disqualified [NOT progressed]
- **SQO**: 254 (1.5%) - became opportunities [PROGRESSED ✅]
- **Won**: 40 (0.2%) - closed deals [PROGRESSED ✅]
- **Lost**: 12 (0.1%) - deal lost [NOT progressed]
- **MQL**: 10 (0.1%) - moved back to marketing [NOT progressed]
- **INQ**: 2 (<0.1%) - moved back to inquiry [NOT progressed]

For Analysis: Create a binary variable where SQL, SQO, and Won = Success (18.0%); all others = Failure (82.0%). Exclude the 394 null values (still in progress).

🏢 ACCOUNT CHARACTERISTICS

`acct target industry` - Industry Sector

Data Type: Text (Categorical)

Missing Values: 45 (0.27%) - excellent data quality ✅

Populated by: Prospect within form fill

5 Unique Values with Distribution:

- **Pharma & BioTech**: 6,343 (38.1%) - drug/biologic manufacturers
- **Non-Life Science**: 5,826 (35.0%) - food, beverage, cosmetics, chemicals
- **Medical Device**: 3,791 (22.8%) - device manufacturers
- **Blood & Biologics**: 458 (2.8%) - blood banks, plasma centers
- **Unknown**: 181 (1.1%)

Why It Matters: Different industries may have different needs for Mx vs Qx. Pharma companies with complex manufacturing might need Mx more than non-life science companies.

`acct manufacturing model` - *Business Model*

Data Type: Text (Categorical)

Missing Values: 101 (0.61%) - excellent data quality ✅

Populated by: AI prompt

19 Unique Values - Top 10:

1. **In-House**: 7,094 (42.6%) - manufacture their own products
2. **Not Enough Info Found**: 2,425 (14.6%)
3. **Services/Consulting**: 1,753 (10.5%) - may not need Mx
4. **CDMO**: 1,105 (6.6%) - Contract Development & Manufacturing
5. **CMO**: 982 (5.9%) - Contract Manufacturing
6. **Original Equipment Manufacturing**: 615 (3.7%)
7. **Software/Technology Development**: 513 (3.1%) - may not need Mx
8. **Distribution/Wholesale**: 446 (2.7%) - may not need Mx
9. **CRO**: 427 (2.6%) - Contract Research (research only)
10. **Research Institute**: 344 (2.1%) - may not need Mx

Key Terminology:

- **CDMO**: Does both R&D and manufacturing for clients (potential for both Mx + Qx?)
- **CMO**: Manufactures products for other companies (strong Mx prospect?)
- **CRO**: Research services only, no manufacturing (better for Qx than Mx?)

Why It Matters: Companies that actually manufacture (In-House, CMO, CDMO) are likely better Mx prospects than service providers. With 99.39% complete data, this is a strong predictor field.

Analysis Consideration: In-House dominates at 42.6%, while Services/Consulting represents 10.5% and Software/Tech 3.1%. Should these non-manufacturing segments even be targeted for Mx?

`acct primary site function` - *What They Manufacture*

Data Type: Text (Categorical)

Missing Values: 60 (0.36%) - excellent data quality ✅

Populated by: AI prompt

78+ Unique Categories - Top 20:

1. **Non-manufacturing organization**: 5,525 (33.2%)
2. **Not Enough Info Found**: 2,550 (15.3%)
3. Specialty Chemical Manufacturing: 656
4. **Oral solid dosage (OSD)**: 539 - pharmaceutical pills
5. **Implantable devices**: 491 - medical devices
6. Food and beverage: 481
7. **In-vitro diagnostic (IVD) kits**: 419 - medical devices
8. **Sterile injectables**: 394 - pharma manufacturing
9. **Small-molecule API**: 360 - pharma ingredients
10. Diagnostic Equipment: 356
11. Nutraceutical manufacturing: 328
12. Lab Instruments: 326
13. **Cell therapies (CAR-T, NK)**: 305 - advanced biologics
14. Drug discovery & development: 300
15. **Biologic API**: 257 - pharma biologics
16. Generics manufacturing: 213
17. Surgical instruments: 209
18. Secondary packaging: 205
19. Cosmetics manufacturing: 186
20. Single-use disposables: 146

Why It Matters: This is critical for identifying which types of facilities are best Mx prospects. With 99.64% complete data, this is a strong predictor field.

Key Insight: The largest single category is "Non-manufacturing organization" at 33.2% (5,525 records). Are these appropriate targets for a Manufacturing solution? This warrants investigation.

Suggested Groupings for Analysis:

- **Active Pharma Manufacturing**: OSD, sterile injectables, API, biologics, cell therapies
- **Medical Device Manufacturing**: Implantable devices, IVD kits, surgical instruments
- **Non-Manufacturing**: Services, consulting, non-manufacturing orgs
- **Other Manufacturing**: Food & beverage, nutraceuticals, cosmetics, chemicals
- **Unknown**: Not Enough Info Found

`acct tier rollup` - *Company Size*

Data Type: Text (Categorical)

Missing Values: 45 (0.27%) - excellent quality ✅

Populated by: Prospect within form fill

4 Unique Values:

- **Medium**: 9,739 (58.5%)
- **Small**: 4,449 (26.7%)
- **Large**: 2,201 (13.2%)
- **Other**: 210 (1.3%)

Why It Matters: Deal size and conversion rates may vary by company size. Mid-size companies might be the sweet spot for Mx.

`acct territory rollup` - *Geographic Region*

Data Type: Text (Categorical)

Missing Values: 132 (0.79%) - excellent quality ✅

Populated by: Prospect within form fill

6 Unique Values:

- **Americas**: 9,550 (57.4%) - North & South America
- **EMEA**: 4,413 (26.5%) - Europe, Middle East, Africa
- **APAC & Oceania**: 2,376 (14.3%) - Asia Pacific & Australia
- **Japan**: 164 (1.0%) - separate territory
- **Unknown**: 6 (<0.1%)
- **Antarctica**: 3 (<0.1%)

Why It Matters: Regional differences in Mx adoption may exist due to market maturity or manufacturing practices.

👤 CONTACT CHARACTERISTICS

`contact/lead title` - *Job Title*

Data Type: Text (Free-form)

Missing Values: 4,862 (29.21%) - 70.79% of records have titles

Populated by: Prospect within form fill or enriched from Zoom Info

~4,000+ Unique Values - highly variable and unstructured

Sample Values: "QA Manager", "VP Quality & Regulatory Affairs", "Quality Systems Manager", "Director of Manufacturing Operations"

What's Included:

- **Seniority indicators**: CEO, Chief, President, VP, Director, Manager, Senior, Lead, Specialist, Analyst, Coordinator, Engineer
- **Functional areas**: Quality, Regulatory, Operations, Manufacturing, Production, IT, Systems, Technology, R&D, Engineering, Compliance

Why This Is Your Most Important Field:
The primary goal is to determine which titles convert best so the sales team knows:

1. Which titles to target for outbound prospecting
2. Which additional contacts to add when receiving inbound leads
3. Whether optimal titles differ between Mx and Qx

Analysis Requirement: You'll need to extract structured information from these free-text titles:

- **Seniority level**: C-Level, VP, Director, Manager, Specialist, Technical, etc.
- **Functional area**: Quality, Regulatory, Operations/Manufacturing, IT/Systems, R&D/Engineering
- **Keyword flags**: has_quality, has_regulatory, has_manufacturing, has_operations, has_manager, has_director, etc.

`contact/lead id` - *Person Identifier*

Data Type: Text (Alphanumeric ID)

Missing Values: 3 (0.02%) - nearly perfect ✅

15,371 Unique Contacts

Format: Salesforce-style ID with "003" prefix (e.g., "0030c00002XPeNGAA1")

Note: 16,644 QALs from 15,371 contacts = 1.08 QALs per contact on average. Some contacts generated multiple leads over time, which could be used for engagement history features.

🎯 PRODUCT INFORMATION

`solution_rollup` - *Product Family (High-Level)*

Data Type: Text (Categorical)

Missing Values: 0 (0%) - perfect quality ✅

Populated by: Logic based on the type of content they download

2 Unique Values: Qx, Mx (Ax filtered out from this dataset)

Distribution:

- **Qx (Quality)**: 12,466 (74.9%) - 19.7% progression rate
- **Mx (Manufacturing)**: 4,178 (25.1%) - 12.7% progression rate

The Core Business Question: Qx has a 55% higher progression rate than Mx (19.7% vs 12.7%). Do different account types and job titles perform better for one product vs. the other?

Analysis Approach: Split your dataset by this field and analyze Mx and Qx separately, then compare findings to understand what drives the performance gap.

`solution` - *Specific Product (Granular)*

Data Type: Text (Categorical)

Missing Values: 0 (0%) - perfect quality ✅

Populated by: Logic based on the type of content they download

14 Unique Values - Top 10:

1. **Qx**: 9,422 (56.6%)
2. **Mx**: 3,246 (19.5%)
3. **Quality**: 1,373 (8.2%)
4. **MISSING/BLANK**: 963 (5.8%)
5. **Logbooks**: 505 (3.0%)
6. **Manufacturing**: 427 (2.6%)
7. **Platform**: 378 (2.3%)
8. **Regulatory**: 113 (0.7%)
9. **Product Development**: 76 (0.5%)
10. **Clinical**: 61 (0.4%)

When to Use: Use `solution_rollup` for high-level Mx vs Qx comparison. Use this field for detailed analysis of specific modules (e.g., does "Logbooks" perform differently than core "Qx"?).

📈 LEAD SOURCE INFORMATION

`Priority` - *Lead Quality/Urgency*

Data Type: Text (Categorical)

Missing Values: 0 (0%) - perfect quality ✅

Populated by: Logic based on the type of content they download

11 Unique Values - Complete Distribution:

1. **Priority 2**: 7,356 (44.2%) - standard priority
2. **Priority 1**: 3,805 (22.9%) - high priority
3. **P1 - Video Demo**: 1,829 (11.0%) - requested demo
4. **P1 - Contact Us**: 1,683 (10.1%) - inbound inquiry
5. **P1 - Webinar Demo**: 874 (5.3%)
6. **No Priority**: 620 (3.7%)
7. **P1 - Website Pricing**: 314 (1.9%)
8. **P1 - Live Demo**: 264 (1.6%)
9. **P1 - MQL Referral**: 45 (0.3%)
10. **Other**: 23 (0.1%)
11. **P1 - Discount Offer**: 2 (<0.1%)

Why It Matters: "P1 -" prefix indicates specific high-intent actions. Demo requests total 2,967 (17.8%) and likely indicate strong conversion potential. This could be a strong predictor.

`last tactic campaign channel` - *Marketing Channel*

Data Type: Text (Categorical)

Missing Values: 0 (0%) - perfect quality ✅

Populated by: Logic based on the type of content they download

12 Unique Values:

1. **Online Ads**: 4,546 (27.3%)
2. **Email**: 3,776 (22.7%)
3. **SEO**: 2,947 (17.7%)
4. **Direct/Inbound**: 1,623 (9.8%)
5. **External Demand Gen**: 1,612 (9.7%)
6. **Directory Listing**: 1,026 (6.2%)
7. **Outbound Prospecting**: 366 (2.2%)
8. **Events**: 357 (2.1%)
9. **Referrals**: 153 (0.9%)
10. **MISSING/BLANK**: 149 (0.9%)
11. **PR/AR/Social**: 86 (0.5%)
12. **Out of Home**: 3 (<0.1%)

Channel Categories:

- **Digital Paid**: Online Ads (27.3%)

- **Digital Organic**: SEO (17.7%), Directory Listing (6.2%)
- **Direct Outreach**: Email (22.7%), Outbound Prospecting (2.2%)
- **Inbound**: Direct/Inbound (9.8%)

Why It Matters: Channel effectiveness analysis and understanding whether certain channels work better for Mx vs Qx.

🔑 IDENTIFIERS & TIME

`QAL ID` - *Lead Identifier*

Data Type: Text (Alphanumeric ID)

Missing Values: 0 (0%)

16,644 Unique Values - 100% unique, serves as primary key

Format: Salesforce-style ID with "a0d" prefix (e.g., "a0dPQ000005WbfuYAC")

Purpose: Unique identifier for each lead; not useful for modeling.

`qal cohort date` - *Qualification Date*

Data Type: Date (YYYY-MM-DD format)

Missing Values: 0 (0%) - perfect quality ✅

Date Range: January 1, 2024 to December 31, 2025 (2 complete years)

Temporal Distribution:

- **2024**: 9,193 QALs (55.2%)
- **2025**: 7,451 QALs (44.8%)

Analysis Potential: Create time-based features like day of week, month, quarter, season. Compare 2024 vs 2025 performance - did Mx improve over time?

📊 DATA QUALITY SUMMARY

Missing Data by Column

Column	Missing	% Missing	Quality
contact/lead title	4,862	29.21%	70.79% complete
next_stage__c	394	2.37%	Still in progress
acct territory rollup	132	0.79%	Excellent
acct manufacturing model	101	0.61%	Excellent
acct primary site function	60	0.36%	Excellent
acct tier rollup	45	0.27%	Excellent
acct target industry	45	0.27%	Excellent
contact/lead id	3	0.02%	Near perfect
All other columns	0	0.00%	Perfect

🎓 GETTING STARTED

Step 1: Load and Explore

1. Load `Mastercontrol QAL Performance.csv` into your analysis tool
2. Verify the statistics match this document
3. Note the strong data quality - excellent completeness across most fields

Step 2: Define Success

Create your target variable: leads that progressed (SQL, SQO, Won) = 1; others = 0
Exclude the 394 records still in QAL stage (null values)

Step 3: Understand the Context

- Why might Mx underperform Qx? (newer product, different buyer personas?)
- What does the 33% "non-manufacturing organization" segment mean for Mx?
- How should missing title data be handled? (treat as separate category)

Step 4: Key Areas to Investigate

Account-Level Questions:

- Which industries show highest Mx conversion? Does this differ from Qx?
- Do companies with actual manufacturing (In-House, CMO, CDMO) convert better for Mx than services/consulting firms?

- What about the 5,525 "non-manufacturing organizations" - should they be targeted for Mx?
- Does company size matter differently for Mx vs Qx?

Title-Level Questions (most value available):

- What seniority levels convert best for each product?
- Which functional areas (Quality, Operations, Regulatory) work better for which product?
- Are there specific title patterns that predict Mx success?
- What is our Mx ICP (Ideal Customer Profile)?

Combined Analysis:

- Which account + title combinations show highest Mx conversion?
- Which account types and titles are good for BOTH products (cross-sell)?
- Are there interaction effects (e.g., Operations Manager works great at In-House manufacturers but not at CROs)?

📚 KEY TERMINOLOGY

Term	Definition
QAL	Quality Accepted Lead - qualified by sales development team
SDR/BDR	Sales/Business Development Representatives who qualify leads
SQL	Sales Qualified Lead - ready for sales team
SQO	Sales Qualified Opportunity - active opportunity created
CDMO	Contract Development & Manufacturing Organization
CMO	Contract Manufacturing Organization
CRO	Contract Research Organization
API	Active Pharmaceutical Ingredient
OSD	Oral Solid Dosage (pills/tablets)
IVD	In-Vitro Diagnostic devices

⚠️ IMPORTANT CONSIDERATIONS

Analysis Notes

- **Two-level analysis needed**: First identify best account types for Mx, then identify best titles within those accounts

- **Success = SQL/SQO/Won**: Don't analyze "Won" alone (only 40 records)
- **18% positive class**: Not severely imbalanced - standard ML techniques work
- **Missing titles (29.21%)**: Don't drop these records; missing might be informative

Business Context

- **Mx is newer**: 4 years vs Qx's 20+ years - still learning optimal customer profiles
- **Performance gap**: Qx outperforms Mx by 55% - this is normal for new products but needs improvement
- **Immediate next stage only**: You're measuring if leads progressed, not if they eventually closed

---

🎯 YOUR DELIVERABLE

Your analysis should provide **clear, actionable recommendations** that answer:

1. **Which account types** should MasterControl target for Mx? (industries, manufacturing models, site functions)
2. **Which job titles** convert best for Mx within those account types?
3. **How do these differ** from Qx's optimal profile?
4. **Which accounts and titles** represent cross-sell opportunities (good for both)?
5. **What's the expected impact** of following your recommendations?

Example Strong Conclusion:

"By targeting Operations Managers and Manufacturing Directors at In-House Pharma companies making sterile injectables or biologics, MasterControl could improve Mx conversion from 12.7% to an estimated 16-18%. These account/title combinations show 2.5x higher conversion than the current average."

Good luck with your analysis! Focus on actionable insights the sales team can implement immediately. The strong data quality gives you a solid foundation for robust findings.